



Efficacy and safety of mitapivat in adults with transfusion-dependent α -thalassaemia or β -thalassaemia (ENERGIZE-T): a double-blind, randomised, multicentre, placebo-controlled, phase 3 trial

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Summary

Background The absence of disease-modifying therapies for patients with α -thalassaemia and oral disease-modifying therapies for patients with β -thalassaemia has been a substantial unmet need in these patients. We assessed the efficacy and safety of mitapivat, an oral allosteric activator of pyruvate kinase, in adults with transfusion-dependent thalassaemia.

Methods ENERGIZE-T is a global, double-blind, randomised, placebo-controlled, phase 3 trial, conducted across 19 countries in North America, Europe, Asia-Pacific, South America, and the Middle East. Patients aged 18 years or older with transfusion-dependent α -thalassaemia or β -thalassaemia were randomly allocated (2:1) with a central interactive response technology system, stratified by geographical region and thalassaemia genotype, to receive 100 mg mitapivat or placebo orally twice a day for 48 weeks. The primary endpoint was transfusion reduction response (TRR), defined as a reduction of at least 50% in transfused red blood cell units with a reduction of at least two units in any consecutive 12-week period until week 48 compared with baseline. Efficacy was analysed in the full analysis set, comprising all randomly allocated patients. Type, severity, and relationship of adverse events and serious adverse events were assessed in patients who received at least one dose of study treatment. This study is registered with ClinicalTrials.gov (NCT04770779) and is active but not recruiting.

Findings Between Nov 30, 2021 and May 2, 2023, 305 patients were screened, of whom 258 were randomly allocated (median age 33·5 years [IQR 27·0–44·0]; 136 [53%] female and 122 [47%] male participants). Of 258 patients allocated, 238 (92%) completed the double-blind treatment period. All patients were required to have a safety follow-up approximately 4 weeks after the final dose of study drug, regardless of completion of the double-blind period or continuation into the open-label extension period. In the full analysis set, TRRs occurred in 52 (30%) of 171 patients in the mitapivat group and 11 (13%) of 87 in the placebo group (adjusted difference 18 percentage points [95% CI 8–27]; two-sided $p=0\cdot0003$). The safety analysis set comprised 172 patients in the mitapivat group (including one patient allocated to the placebo group who received one dose of mitapivat in error) and 85 in the placebo group. Adverse events were reported in 155 (90%) patients treated with mitapivat and 71 (84%) treated with placebo; the most common events with mitapivat were headache, upper respiratory tract infection, initial insomnia, diarrhoea, and fatigue. Serious adverse events were reported in 19 (11%) patients treated with mitapivat and 13 (15%) treated with placebo. Ten (6%) patients who received mitapivat and one (1%) who received placebo discontinued study treatment due to adverse events. No deaths were reported.

Interpretation Mitapivat significantly reduced the transfusion burden and was generally well tolerated, showing a favourable benefit–risk profile. These findings support mitapivat as the first oral disease-modifying therapy for adults with transfusion-dependent α -thalassaemia or β -thalassaemia, providing a new treatment option to reduce transfusion burden in this patient population.

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Introduction

Thalassaemia is a diverse group of inherited blood disorders of haemoglobin synthesis.¹ In thalassaemia,

imbalances in globin chain production during erythropoiesis result in increased oxidative stress and damage, leading to ineffective erythropoiesis, haemolysis,

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Research in context

Evidence before this study

Patients with transfusion-dependent thalassaemia require lifelong regular red blood cell transfusions to maintain haemoglobin concentrations and suppress ineffective erythropoiesis. Regular transfusions are associated with health complications and substantial patient burden. We searched PubMed for publications in English, from database inception to Oct 31, 2021, on clinical trials (phase 1–3) in adults, and guidelines on drug treatments for thalassaemia, using the terms “alpha-thalassemia” [Title] OR “beta-thalassemia” [Title] OR “thalassemia” [Title] AND “treatment” [Title/Abstract]. Studies focusing only on non-transfusion-dependent thalassaemia were excluded. 18 publications were identified, of which one was excluded since it did not meet the search criteria for transfusion-dependent thalassaemia. The relevant trials covered a range of pharmacological interventions in patients with thalassaemia, including managing iron overload with iron chelators (eg, deferasirox), transforming growth factor- β ligand modulators, and disease-modifying agents (including subcutaneous luspatercept, which was approved for transfusion-dependent β -thalassaemia in the USA in 2019 and for transfusion-dependent and non-transfusion-dependent β -thalassaemia in Europe in 2020 and 2023, respectively). At the time of the literature search, a few gene therapy approaches had been evaluated in β -thalassaemia, including betibeglogene autotemcel, which was later approved. However,

as of the search date, there were no disease-modifying therapies approved for α -thalassaemia, and there are no oral disease-modifying therapies approved for β -thalassaemia.

Added value of this study

To our knowledge, ENERGIZE-T is the first randomised, phase 3 study to assess a pyruvate kinase activator (mitapivat) in adults with transfusion-dependent thalassaemia (including both α -thalassaemia and β -thalassaemia). The results of this study have established pyruvate kinase as a treatment target in patients with transfusion-dependent thalassaemia, showing significant reductions in transfusion burden in a globally representative population from 19 countries in North America, Europe, Asia-Pacific, South America, and the Middle East. Mitapivat was generally well tolerated and had a favourable benefit-risk profile. This study is also the first to show efficacy of an orally administered disease-modifying therapy in transfusion-dependent β -thalassaemia and the first to show efficacy of a disease-modifying therapy in transfusion-dependent α -thalassaemia.

Implications of all the available evidence

The findings of this global, randomised, phase 3 study support the use of mitapivat in patients with transfusion-dependent α -thalassaemia or β -thalassaemia, and offer the first oral disease-modifying therapy for this disease.

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See Online for appendix

and dysregulated iron homeostasis; if left untreated, this condition can lead to serious complications and death.^{2,3} The degree of anaemia in thalassaemia is dependent on the magnitude of the imbalance between α -globin and β -globin chain production, and translates into a range of red blood cell transfusion needs across all genotypes.^{4,5}

Regular red blood cell transfusions are required for transfusion-dependent thalassaemia, while no or occasional transfusions might be required for non-transfusion-dependent thalassaemia.⁶ The goal of red blood cell transfusion therapy is to reduce ineffective erythropoiesis, haemolysis, anaemia, and their associated complications, such as primary iron overload, growth delays, skeletal abnormalities, leg ulcers, and extramedullary haematopoiesis.^{2,7} The recommended target pretransfusion haemoglobin concentration for most patients with transfusion-dependent thalassaemia is 9.5–10.5 g/dL to promote normal growth and physical activity, suppress ineffective erythropoiesis,⁷ and reduce thalassaemia-related mortality.⁸ However, patients with transfusion-dependent thalassaemia can still have complications due to the disease physiopathology itself, particularly if suboptimally managed.⁷ Furthermore, red blood cell transfusion therapy contributes to secondary morbidity and increases health-care resource utilisation.^{1,3,9–11}

The treatment-related burden of transfusion-dependent thalassaemia can cause substantial patient-reported effects, resulting in reduced quality of life in social and family life, work or education, and mental health.¹² Medical claims database studies show that transfusion-dependent thalassaemia is associated with high health-care resource utilisation, driven by iron-chelation therapy, red blood cell transfusions, comorbidities, and hospitalisations.^{11,13,14} Thus, lifelong care for many patients with transfusion-dependent thalassaemia remains resource intensive, costly, and with treatment options limited by patient eligibility, genotype, transfusion requirement, and access.^{2,5,7,10,13–16}

Iron overload secondary to red blood cell transfusion and disease-related primary iron overload are the targets of corrective iron-chelation therapy.^{1–3,6} Despite use of chelation, patients who receive regular transfusions can develop complications associated with secondary iron overload, including hepatic, cardiac, and endocrine manifestations, and subsequent organ damage.⁷ Additionally, iron-chelating agents are associated with toxicity, which requires monitoring.¹⁷

Other treatment options include a subcutaneous erythroid maturation agent (luspatercept,¹⁸ approved for transfusion-dependent β -thalassaemia in the USA in 2019 and for transfusion-dependent and non-transfusion-dependent β -thalassaemia in Europe in 2020 and 2023,

respectively); allogeneic haematopoietic stem-cell transplantation;² and, for transfusion-dependent β -thalassaemia, gene therapy through CRISPR–Cas9 gene editing (exagamglogene autotemcel;¹⁹ approved in the USA and Europe in 2024) or gene addition (betibeglogene autotemcel;²⁰ approved in the USA in 2022).

In thalassaemic red blood cells, ATP production is insufficient to support the increased energy demand for clearing globin aggregates, ameliorating oxidative stress, and repairing oxidative damage.^{21–24} Pyruvate kinase (PK) activation increases ATP production to potentially address the increased demand for ATP, which can reduce ineffective erythropoiesis and haemolysis, leading to improvements in anaemia and transfusion burden.^{23,25} Mitapivat—a first-in-class oral, allosteric activator of the red blood cell-specific isoform of PK (PKR) and PKM2—is approved in adults with α -thalassaemia or β -thalassaemia, regardless of transfusion needs, in the USA, EU, Saudi Arabia, and United Arab Emirates.^{26–28} In a phase 2 study (NCT03692052) in adult patients with non-transfusion-dependent α -thalassaemia or

β -thalassaemia, mitapivat was associated with improvements in haemoglobin levels and markers of ineffective erythropoiesis and haemolysis, consistent with mitapivat's mechanism of PK activation.²⁹ In ENERGIZE (NCT04770753), a phase 3 trial involving patients with non-transfusion-dependent thalassaemia, including both α -thalassaemia and β -thalassaemia, mitapivat significantly improved haemoglobin concentration and fatigue compared with placebo, and was generally well tolerated with a low treatment discontinuation rate.³⁰

In this Article, we report findings from the ENERGIZE-T³¹ clinical trial, which investigated the efficacy and safety of oral mitapivat versus placebo in adult patients with transfusion-dependent thalassaemia.

Methods

Study design

ENERGIZE-T is a double-blind, randomised, placebo-controlled, superiority, phase 3 trial conducted at 81 sites across 19 countries in North America, Europe, Asia-Pacific, South America, and the Middle East (appendix pp 3–8, 17). The study consists of a 48-week double-blind period followed by a 5-year open-label extension period, which is ongoing. This Article reports on the results from the 48-week double-blind period. This study was conducted in accordance with the Declaration of Helsinki and International Council for Harmonization Good Clinical Practice Guidelines. All local regulatory requirements were followed. The study protocol and informed consent forms were approved by institutional review boards or ethics committees at each study site. Internal and external review committees monitored the safety of patients throughout the study. All patients provided written informed consent. The study protocol is available in the appendix (pp 26–119), along with detailed information on protocol deviations (p 25). The statistical analysis plan is available in the appendix (pp 117–162). An independent data monitoring committee reviewed the data and provided ongoing monitoring of patient safety during the double-blind period. The trial is registered at ClinicalTrials.gov (NCT04770779).

Patients

Patients were screened for eligibility within an 8-week window. Eligible patients were aged 18 years or older with a documented diagnosis of thalassaemia (β -thalassaemia with or without α -globin gene mutations, HbE/ β -thalassaemia, or α -thalassaemia/HbH disease) based on DNA analysis. All eligible patients were transfusion dependent, defined per study protocol as six to 20 red blood cell units transfused and up to a 6-week transfusion-free period during the 24-week period before randomisation. Exclusion criteria included receiving treatment with luspatercept or a haematopoietic-stimulating agent (last dose must have been administered ≥ 36 weeks before randomisation)

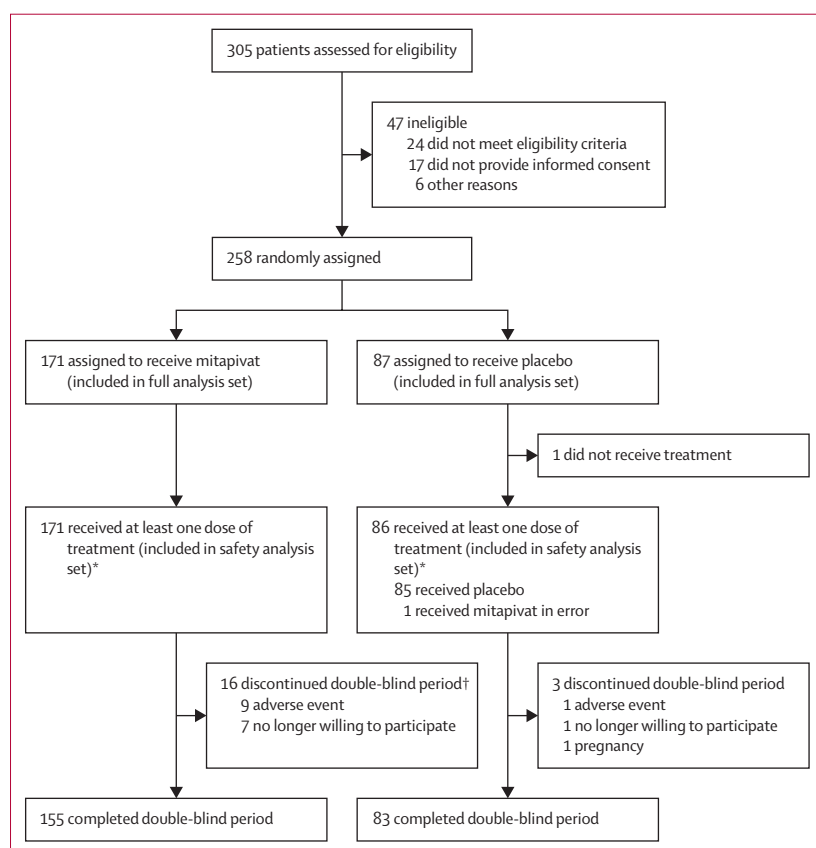


Figure 1: Trial profile

*The safety analysis set for mitapivat (n=172) comprised all patients who received at least one dose of mitapivat, including one participant randomly allocated to placebo who mistakenly received a dose of mitapivat in the double-blind period; the safety analysis set for placebo (n=85) comprised all patients who received at least one dose of placebo, excluding the patient who mistakenly received one dose of mitapivat. †One patient discontinued study treatment due to an adverse event (not included in figure) and later discontinued from the double-blind period because they were no longer willing to participate.

and previous exposure to gene therapy or previous stem-cell transplantation. Transfusions, iron-chelation therapy, and a stable dose of hydroxyurea were permitted as clinically indicated. Full eligibility criteria are shown in the appendix (pp 9–10).

Randomisation and masking

Eligible patients were randomly assigned in a 2:1 ratio to the mitapivat group or placebo group in a parallel group design, using a central randomisation scheme provided by an interactive response technology system. Enrolling sites did not have access to the random allocation sequence. Randomisation was stratified by geographical region (North America and Europe, Asia–Pacific, and rest of world) and by thalassaemia genotype ($\beta^0\beta^0$ versus non- $\beta^0\beta^0$ [non- $\beta^0\beta^0$ included HbE/ β -thalassaemia and α -thalassaemia/HbH disease]; appendix p 74). Treatment randomisation was stratified by geographical region to account for regional differences in transfusion practices. All trial participants, investigators, clinical study centre personnel, pharmacists, and the sponsor were masked to the treatment assignment. Mitapivat and matched placebo were packaged and labelled identically to maintain masking.

Procedures

A pretransfusion haemoglobin threshold was ascertained for each patient based on transfusion history, and defined as the mean of all documented pretransfusion haemoglobin concentration values recorded for the red blood cell transfusions administered during the 24 weeks before randomisation. During the study, if haemoglobin concentration was 1 g/dL or more above the pretransfusion threshold and within the upper limit of normal (by sex), investigators had the option to delay transfusion or to reduce the number of red blood cell units to be transfused. Guidance on red blood cell transfusion modification is available in the study protocol (appendix pp 78–79).

Patients in the mitapivat group received oral mitapivat (100 mg twice per day) and those in the placebo group received matching placebo (twice per day) for 48 weeks, with study visits approximately every 4 weeks, at which efficacy and safety assessments were conducted, as specified in the protocol (appendix pp 43–48).

Adverse events were assessed continuously over the 48-week double-blind period. Number and percentage of patients with adverse events are reported; multiple adverse events within a Preferred Term are counted only once in that Preferred Term. For patients with multiple occurrences of an adverse event, the event with the worst grade based on the Common Terminology Criteria for Adverse Events (version 4.03) is included in the summary. Causality and seriousness were assigned by study site investigators while masked to treatment allocation. All patients were required to have a safety follow-up approximately 4 weeks after the last dose of study drug,

	Mitapivat group (n=171)	Placebo group (n=87)
Age, years		
Median (IQR)	33 (27–45)	34 (26–43)
Range	18–67	20–53
Sex		
Female	93 (54%)	43 (49%)
Male	78 (46%)	44 (51%)
Race		
White	99 (58%)	56 (64%)
Asian	56 (33%)	22 (25%)
Black or African American	1 (1%)	1 (1%)
Multiracial	2 (1%)	0
Unknown	7 (4%)	3 (3%)
Not reported	6 (4%)	5 (6%)
Ethnicity		
Hispanic or Latino	5 (3%)	3 (3%)
Not Hispanic or Latino	158 (92%)	79 (91%)
Not reported	8 (5%)	5 (6%)
Thalassaemia genotype		
Non- $\beta^0\beta^0$	96 (56%)	48 (55%)
$\beta^0\beta^0$	75 (44%)	39 (45%)
24-week transfusion burden*		
≤12 red blood cell units	54 (32%)	21 (24%)
>12 red blood cell units	117 (68%)	66 (76%)
Pretransfusion haemoglobin threshold, g/dL†		
Mean (SD)	8.9 (1.2)	8.8 (1.1)
Median (IQR)	9.0 (8.1–9.6)	8.9 (8.0–9.7)
Range	5.1–11.8	5.1–10.9
Splenectomy status‡		
No	79 (46%)	38 (44%)
Yes	92 (54%)	49 (56%)
Receipt of iron chelation within 1 year before randomisation‡		
No	6 (4%)	0
Yes	165 (96%)	87 (100%)
Hydroxyurea use§		
No	164 (96%)	84 (97%)
Yes	7 (4%)	3 (3%)
Mean ferritin concentration, µg/L¶	2020.0 (2290.9), n=161	1928.7 (1862.1), n=83
Mean hepatic iron by MRI, mg/g¶	5.7 (4.9), n=133	5.7 (4.7), n=74

Data are median (IQR), n (%), or mean (SD), except where otherwise specified.
 *24-week baseline transfusion burden is the total number of red blood cell units transfused during the 24-week period (168 days) before the reference date (ie, the start of study treatment, if received, or the randomisation date otherwise).
 †Defined as the mean of all documented pretransfusion haemoglobin concentration values recorded for the red blood cell transfusions administered during the 24-week period before randomisation. ‡As recorded in thalassaemia disease history electronic case report forms. §Patients treated with hydroxyurea were required to be on a stable dose for at least 16 weeks before randomisation. ¶Baseline was defined as the last assessment before the last red blood cell transfusion before randomisation.

Table 1: Baseline demographics, disease characteristics, and laboratory values of patients in the full analysis set

regardless of whether they discontinued before the end of the double-blind period, did not enter into the open-label extension period, or continued into the open-label extension period.

Outcomes

The primary endpoint was transfusion reduction response (TRR), defined as reduction of at least 50% in transfused red blood cell units with a reduction of at least two units of transfused red blood cells in any consecutive 12-week period until week 48 compared with baseline. The key secondary endpoints were TRR2, defined as a reduction of at least 50% in transfused red blood cell units in any consecutive 24-week period until week 48, compared with baseline; TRR3, defined as a reduction of at least 33% in transfused red blood cell units from week 13 to week 48, compared with baseline; and TRR4, defined as a reduction of at least 50% in transfused red blood cell units from week 13 to week 48, compared with baseline. A secondary endpoint measuring transfusion burden, also reported

herein, was transfusion independence, defined as the patient being transfusion free for at least 8 consecutive weeks within the 48-week double-blind period.

Additional prespecified secondary endpoints are described in the appendix (pp 15, 32–33) but not reported in this Article, and will be reported in a future publication. Secondary endpoints were change from baseline in transfused red blood cell units from week 13 to week 48; change from baseline in iron, serum ferritin, total iron binding capacity, and transferrin saturation to week 48; and plasma or blood concentrations and pharmacokinetic parameters of mitapivat and pharmacodynamic parameters, including ATP and 2,3-diphosphoglycerate.

Statistical analysis

A sample size of 240 patients randomly allocated in a 2:1 ratio to mitapivat or placebo was planned to provide 95% power to detect a treatment effect of TRR rate increase from 12·5% in the placebo group to 33·7% in the mitapivat group at a two-sided type I error rate (α) of 0·05, based on Fisher's exact test. The conservative assumptions for the TRR rate were informed by the results of a phase 3 clinical trial in patients with transfusion-dependent β -thalassaemia, in which 12·5% was the upper bound of the 95% CI for the TRR rate in the placebo group and 33·7% was the lower bound of the 95% CI for the TRR rate for the active treatment group.¹⁸ Overall type I error was maintained at or below a two-sided significance level of 0·05 based on a graphical gatekeeping approach.³² The primary endpoint was tested at a two-sided α level of 0·05. Then first two key secondary endpoints (TRR2 and TRR3) were to be tested at an α -level of 0·018 and 0·032, respectively, if the null hypothesis for the primary endpoint was rejected. The final key secondary endpoint (TRR4) was to be tested at an α level of 0·018 (if only TRR2 was rejected), 0·032 (if only TRR3 was rejected), or 0·05 (if both TRR2 and TRR3 were rejected). For each TRR endpoint, baseline transfusion burden was standardised to match the post-baseline assessment duration.

Efficacy analyses were conducted on the full analysis set, defined as all patients who were randomly allocated, and included all data from the double-blind period. Safety analyses were conducted on the safety analysis set, defined as all patients who received at least one dose of study treatment. Data are reported until the end of the double-blind period, defined as the end of safety follow-up for patients who discontinued study treatment during the double-blind period, or before the start of study treatment in the open-label extension period for patients who continued onto the open-label extension period.

The primary endpoint, key secondary efficacy endpoints, and the secondary efficacy endpoint of transfusion independence were analysed using the Mantel–Haenszel stratum-weighted method adjusting for randomisation stratification factors. Additional details are provided in the appendix (pp 11–12). Patients withdrawn from the study before week 12 (day 85), and

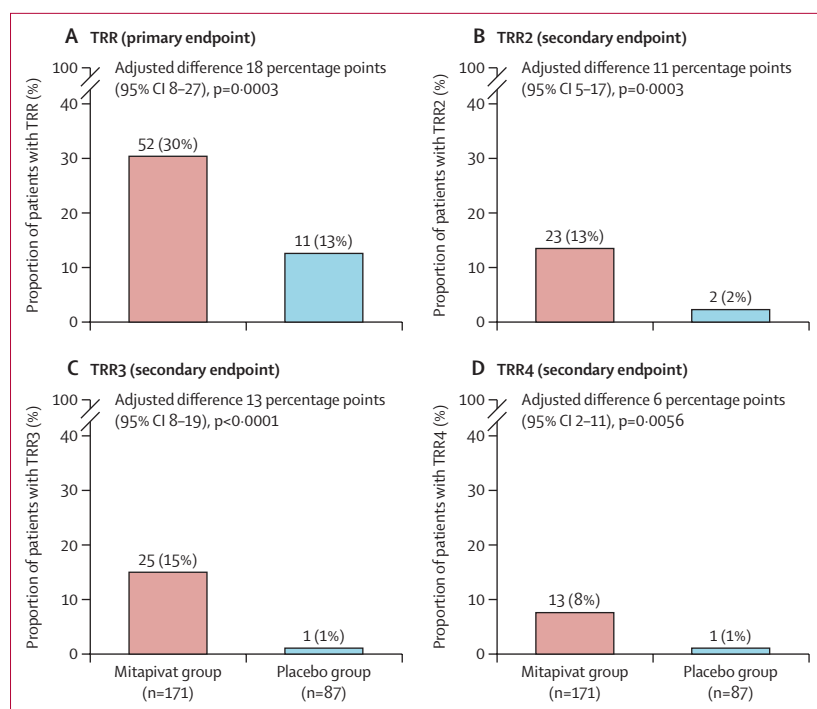


Figure 2: Primary and secondary TRR endpoints

(A) TRR, defined as reduction of at least 50% in transfused red blood cell units with a reduction of at least two units of transfused red blood cells in any consecutive 12-week period until week 48 compared with baseline; patients withdrawn from the study before week 12 (day 85) were considered non-responders. (B) TRR2, defined as at least a 50% reduction in transfused red blood cell units in any consecutive 24-week period until week 48 compared with baseline; patients withdrawn from the study before week 24 (day 169) were considered non-responders. (C) TRR3, defined as at least a 33% reduction in transfused red blood cell units between week 13 and week 48 compared with baseline; patients withdrawn from the study before week 48 were considered non-responders. (D) TRR4, defined as at least a 50% reduction in transfused red blood cell units between week 13 and week 48 compared with baseline; patients withdrawn from the study before week 48 were considered non-responders. For TRR, baseline transfusion burden was standardised to 12 weeks (24-week baseline transfusion burden $\times$ 12 / 24). For TRR2, 24-week baseline transfusion burden was defined as the total number of red blood cell units transfused during the 24-week period (168 days) before the reference date (ie, the start of study treatment, if received, or the randomisation date otherwise). For TRR3 and TRR4, baseline transfusion burden was standardised to 36 weeks (24-week baseline transfusion burden $\times$ 36 / 24). TRR=transfusion response rate.

week 24 (day 169) were considered non-responders for TRR and TRR2, respectively. Patients withdrawn during the double-blind period were considered non-responders for TRR3 and TRR4. Reduction (%) in red blood cell transfusion burden from week 13 to week 48 from the baseline transfusion burden standardised to 36 weeks was analysed using ANCOVA with treatment group as the independent variable, and covariates for randomisation stratification factors and baseline transfusion burden standardised to 36 weeks. Baseline transfusion burden was collected as the total number of red blood cell units transfused during the 24-week period (168 days) before the start of study treatment, for patients randomly allocated and treated, or before the randomisation date otherwise. The baseline transfusion burden was standardised to 12 weeks for the analysis of TRR, and 36 weeks for TRR3 and TRR4. SAS software (version 9.4) was used for all statistical analyses.

Role of the funding source

The funder of the study (Agiros Pharmaceuticals, Inc.) had a role in the study design, data acquisition, data analysis, data interpretation, writing of the report (including medical writing assistance support), and the decision to submit for publication.

Results

Of 305 patients screened between Nov 30, 2021 and May 2, 2023, 258 were randomly assigned to receive mitapivat (n=171) or placebo (n=87). Of those randomly allocated, 238 patients completed study treatment in the double-blind period (155 in the mitapivat group and 83 in the placebo group; figure 1). No patients were lost to follow-up during this period. During the double-blind period, the duration of exposure to treatment was a median of 48.1 weeks (IQR 47.3–48.9; range 0.3–59.9) for mitapivat and 48.0 weeks (IQR 47.1–48.7; range 14.3–53.1) for placebo.

Demographics and baseline characteristics are shown in table 1. The median age of patients at baseline was 33.5 years (IQR 27.0–44.0; range 18–67), and 136 (53%) of 258 patients were female and 122 (47%) were male. 144 (56%) patients had a non- $\beta^0\beta^0$ genotype (appendix p 16).

There were no imbalances in key baseline disease characteristics deemed to affect the interpretation of results. 117 (68%) of 171 patients in the mitapivat group and 66 (76%) of 87 in the placebo group had a 24-week baseline transfusion burden of more than 12 red blood cell units. The mean pretransfusion haemoglobin threshold was 8.85 g/dL for patients in the mitapivat group and 8.82 g/dL for patients in the placebo group; 165 (96%) in the mitapivat group and 87 (100%) in the placebo group had received iron-chelation therapy within 1 year before randomisation, and 92 (54%) and 49 (56%), respectively, had undergone splenectomy. More than 10% of patients in each treatment group reported medical

history events of osteoporosis, cholecystectomy, hepatic disease, hypogonadism, cholelithiasis, and hypothyroidism (appendix p 18).

In the full analysis set, 52 (30%) of 171 patients in the mitapivat group and 11 (13%) of 87 in the placebo group had a TRR (adjusted difference 18 percentage points [95% CI 8–27]; two-sided $p=0.0003$; figure 2).

Mitapivat showed further clinical improvements in reduction of transfusion burden compared with placebo, as measured by the three key secondary TRR endpoints: TRR2 (23 [13%] patients in the mitapivat group vs two [2%] in the placebo group; adjusted difference 11 percentage points [95% CI 5–17], $p=0.0003$); TRR3 (25 [15%] vs one [1%]; 13 percentage points [8–19], $p<0.0001$); and TRR4 (13 [8%] vs one [1%]; 6 percentage points [2–11], $p=0.0056$; figure 2).

For the primary and three key secondary endpoints, subgroup analyses showed that the effects were not driven by any prespecified individual subgroup (appendix pp 13–14). In addition, a descriptive post-hoc analysis in patients with α -thalassaemia (n=12) was conducted. Baseline demographics and disease characteristics for these patients are summarised in the appendix (p 20). The results from this analysis suggest that the effect of

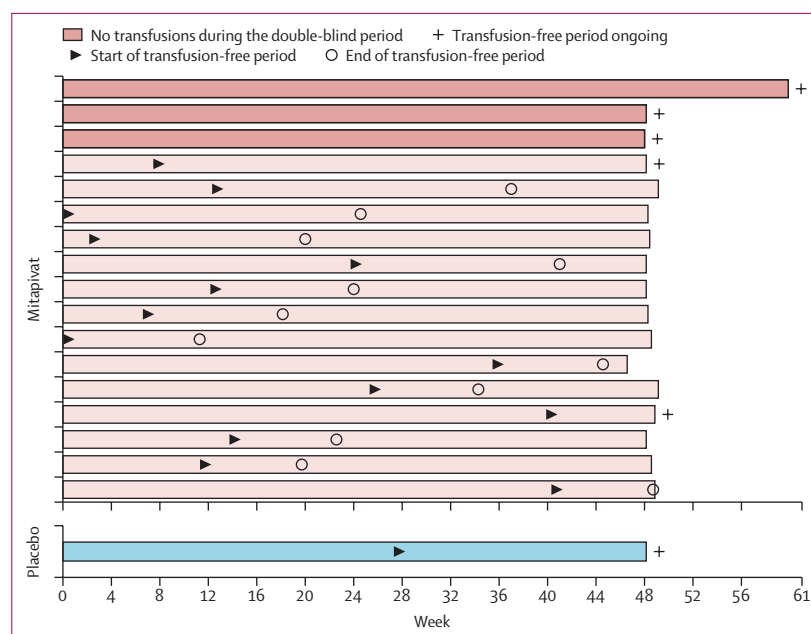


Figure 3: Swimlane plot for duration of transfusion independence for patients who had transfusion independence

Bars are ordered by longest duration of transfusion-free period. Transfusion independence was defined as being transfusion free for at least 8 consecutive weeks until week 48 in the double-blind period. For patients who had transfusion independence, duration of transfusion independence was the number of weeks in the longest transfusion-free period starting on or after the reference date (ie, the start of study treatment, if received, or the randomisation date otherwise), until week 48 in the double-blind period. A transfusion-free period was considered to start on the day after a red blood cell transfusion and end the day before the next red blood cell transfusion. For patients who did not receive any red blood cell transfusions, the start of the transfusion-free period was the reference date. For patients who either did not receive any red blood cell transfusions or received a red blood cell transfusion and no subsequent red blood cell transfusions, the end of the transfusion-free period was censored at the end of the double-blind treatment period.

	Mitapivat group (n=172)	Placebo group (n=85)
Any adverse event	155 (90%)	71 (84%)
Grade ≥3	32 (19%)	12 (14%)
Treatment-related adverse events	65 (38%)	16 (19%)
Grade ≥3	13 (8%)*	1 (1%)†
Serious adverse events	19 (11%)‡	13 (15%)§
Serious treatment-related adverse events	4 (2%)¶	1 (1%)
Adverse events leading to discontinuation of study drug**	10 (6%)††	1 (1%)‡‡
Adverse events leading to dose reduction	20 (12%)	2 (2%)
Adverse events leading to interruption of study drug	13 (8%)	5 (6%)
Adverse events leading to death	0	0
Most frequently reported adverse events of any grade§§		
Headache		
Any grade	46 (27%)	10 (12%)
Grade ≥3	0	0
Upper respiratory tract infection		
Any grade	27 (16%)	14 (16%)
Grade ≥3	0	0
Initial insomnia		
Any grade	24 (14%)	4 (5%)
Grade ≥3	3 (2%)	0
Diarrhoea		
Any grade	19 (11%)	7 (8%)
Grade ≥3	0	0
Fatigue		
Any grade	18 (10%)	2 (2%)
Grade ≥3	0	0

Data are n (%) of patients with at least one event. Adverse events were collected throughout the study and classified using Common Terminology Criteria for Adverse Events (version 4.03). *Events of insomnia (n=3); concurrent increased alanine aminotransferase and increased aspartate aminotransferase (n=2); abdominal distention, increased alanine aminotransferase, hypertransaminasaemia, increased blood pressure, lower respiratory tract infection, supraventricular arrhythmia, supraventricular tachycardia, and renal mass (all n=1). †Increased blood creatine phosphokinase. ‡Gastroenteritis (n=2); pneumonia, COVID-19 pneumonia, cellulitis, dengue fever, influenza, lower respiratory tract infection, hypersplenism, mesenteric lymphadenitis, pancytopenia, cholecystitis, acute cholecystitis, supraventricular arrhythmia, supraventricular tachycardia, radius fracture, proctitis, asthenia, hepatic cancer, dizziness, renal mass, and ruptured ovarian cyst (all n=1). §Pneumonia (n=2); viral infection, splenic haematoma, cholecystitis, acute cholecystitis, acute cholangitis, arrhythmia, left ventricular dysfunction, infusion-related reaction, cataract, increased blood creatine phosphokinase, limb deformity, spontaneous abortion, and pulmonary hypertension (all n=1). ¶Lower respiratory tract infection, renal mass, supraventricular arrhythmia, supraventricular tachycardia. ||Increased blood creatine phosphokinase. **Discontinuation due to an adverse event could be made at the discretion of the investigator, the patient, or a combination of both. ††Diarrhoea, oral paraesthesia, concurrent anxiety and insomnia, initial insomnia, supraventricular tachycardia, fatigue, hypertransaminasaemia, hepatitis C, hepatic cancer, and renal mass (all n=1). ‡‡Increased creatine phosphokinase. §§Occurring in ≥10% of either treatment group, summarised in order of decreasing frequency of patients with events based on the frequencies observed in any grade for the mitapivat group.

Table 2: Adverse events in the safety analysis set

mitapivat in this subgroup is consistent with the overall cohort (appendix pp 15–16).

A higher proportion of patients in the mitapivat group (17 [10%] of 171) than the placebo group (one [1%] of 87) had protocol-defined transfusion independence (adjusted difference 9 percentage points [95% CI 4–14]) including three patients in the mitapivat group who did not receive any transfusions during the 48-week double-blind period (figure 3). Among the 17 patients in the mitapivat group who had transfusion independence, mean haemoglobin concentration during transfusion-free periods was maintained at 1.4 g/dL above their mean pretransfusion haemoglobin threshold of 8.1 g/dL, and six of these patients had a $\beta^0\beta^0$ genotype.

In the safety analysis set, 155 (90%) of 172 patients who received mitapivat and 71 (84%) of 85 who received placebo had adverse events during the 48-week double-blind period (table 2). The most commonly reported adverse events of any grade (occurring in ≥10% of patients in either treatment group) were headache, upper respiratory tract infection, initial insomnia, diarrhoea, and fatigue. Adverse events of grade 3 or higher were reported in 32 (19%) patients who received mitapivat and in 12 (14%) who received placebo. Serious adverse events were reported in 19 (11%) patients who received mitapivat and 13 (15%) patients who received placebo, among which four events in the mitapivat group and one in the placebo group were considered to be related to treatment. Ten (6%) patients who received mitapivat and one (1%) who received placebo discontinued treatment due to adverse events. Dose interruptions due to adverse events occurred in 13 (8%) patients who received mitapivat and five (6%) who received placebo. Additionally, 20 (12%) patients who received mitapivat and two (2%) who received placebo had adverse events that led to dose reduction. There were no deaths in either group.

Discussion

ENERGIZE-T showed the efficacy of mitapivat to reduce transfusion burden in adult patients with α -thalassaemia or β -thalassaemia who are transfusion dependent. Patient baseline disease characteristics were globally representative of a real-world population. In general, randomly allocated patients had a high disease burden, as evidenced by considerable transfusion burden, high use of iron chelators, and elevated hepatic iron concentrations.

Regular red blood cell transfusions remain the cornerstone of treatment for patients with thalassaemia, but are associated with a substantial treatment burden, loss of time to transfusion-related activities, health risks, and high utilisation of health-care resources.^{5,7,11–16} Oral disease-modifying therapies that reduce transfusion burden have the potential to improve patient outcomes and health-care utilisation. In the double-blind period of the current study, statistically significant, clinically

meaningful, and durable reductions in transfusion burden were observed in the mitapivat group compared with the placebo group. An adjusted difference of 18 percentage points was observed between mitapivat and placebo for the primary TRR endpoint, which required a reduction of at least 50% in transfused red blood cell units with a reduction of at least two units of transfused red blood cells in any consecutive 12-week period until week 48 compared with baseline. Some patients might have had a reduction in transfusion burden without meeting the threshold; the 2025 Thalassaemia International Federation guidelines suggest that a reduction of at least 20% in transfusion burden is clinically meaningful.⁷

Mitapivat was generally well tolerated, with a low treatment discontinuation rate. The proportions of patients with serious adverse events and adverse events were similar across treatment groups. Among the patients who received mitapivat and had a treatment-related adverse event leading to treatment discontinuation, one patient had an event of grade 3 hypertransaminasaemia, which improved after mitapivat was stopped. Monitoring of liver test results before starting mitapivat treatment to establish a baseline and then monthly during the first 6 months of mitapivat treatment in patients with thalassaemia is required.²⁸

Limitations of the current study were that it was not designed or powered to evaluate predictors of response to treatment or to make statistical comparisons across subgroups based on point estimates, given the smaller sample sizes within subgroups and inherent variability reflected in the 95% CIs. It could be hypothesised that patients with more severe genotypes and less residual erythropoiesis might be less likely to have a robust response to disease-modifying agents including mitapivat. However, patients with $\beta^0\beta^0$ genotypes were observed among those who responded across all of the increasingly stringent measures of transfusion burden reduction in this study, including the endpoint of transfusion independence for at least 8 weeks. Further studies would be required to better understand the benefits of mitapivat in specific patient subgroups and to identify any potential modulators of response, such as primary and secondary genetic modifiers. Patients receiving luspatercept or haematopoietic-stimulating agents, or who had previous exposure to gene therapies or bone marrow transplant, were excluded from participating. As such, no efficacy or safety data are available for combination approaches or for patients with failed engraftment following therapy. Levels of iron-related markers can fluctuate due to factors beyond iron overload and can be influenced by recent transfusions. Accurate assessment of iron status over short intervals remains challenging; therefore, markers of iron status will continue to be monitored during the long-term extension phase of ENERGIZE-T.

This is the first phase 3, placebo-controlled trial involving adult patients across transfusion-dependent α -thalassaemia

or β -thalassaemia. In ENERGIZE-T, mitapivat significantly reduced transfusion burden, with durability of response up to 36 weeks during the 48-week double-blind period, and was generally well tolerated. These data support mitapivat as a new oral disease-modifying therapy for adult patients with transfusion-dependent thalassaemia. Taken together with the positive findings of the ENERGIZE trial, which evaluated the efficacy of mitapivat in patients with non-transfusion-dependent α -thalassaemia and β -thalassaemia,³⁰ these data show the efficacy of mitapivat, further supporting it as the first oral disease-modifying therapy for patients with this condition regardless of genotype or transfusion burden.

Contributors

MDC, SS, AT, HA-S, JHE, SG, KMM, KHKM, and KO designed the study and protocol. AT, HA-S, ABA, JLD, JHE, EF, PG, SG, AG, KMM, KO, JBP, EJvB, VV, KHKM, and AK collected the data. JHE, EF, SG, GMP, KO, and HS analysed the data. All authors contributed to the drafting, review, and revision of this manuscript, and vouch for adherence of the study to the protocol and the accuracy of the results. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

MDC reports honoraria from Agios Pharmaceuticals. SS reports research funding from Agios Pharmaceuticals, Bristol Myers Squibb (Celgene), Forma/Novo Nordisk, and Regeneron; consultancy or advisory board fees from Agios Pharmaceuticals, Bluebird Bio, Bristol Myers Squibb (Celgene), Chiesi, Fulcrum, and Vertex Pharmaceuticals; participation on a steering committee for CRISPR Therapeutics/Vertex; and honoraria from CCO, PER, and Plexus. AT reports consultancy fees from Agios Pharmaceuticals, Bristol Myers Squibb (Celgene), Novo Nordisk, Pharmacosmos, and Roche; and research funding from Agios Pharmaceuticals, Bristol Myers Squibb (Celgene), Novo Nordisk, Pharmacosmos, and Roche. HA-S reports consultancy fees from Agios Pharmaceuticals, Alnylam, Alpine, Amgen, Argenx, Novartis, Pharmacosmos, Sanofi, and Sobi; and research funding from Agios Pharmaceuticals, Amgen, Novartis, Sobi, and Vadderis. ABA reports honoraria and consultancy fees from Novo Nordisk, Pfizer and Takeda. DB reports honoraria from Agios Pharmaceuticals, Bristol Myers Squibb (Celgene), Novartis, Pfizer, Sobi, and Vertex Pharmaceuticals; and payment for advisory boards from Agios Pharmaceuticals, Alexion, Chiesi, Novartis, Novo Nordisk, Pfizer, Roche, Samsung Bioepis, Sobi, and Vertex Pharmaceuticals. GC reports consultancy or advisory board fees from Agios Pharmaceuticals, Novo Nordisk, Pfizer, and Theravia. TC reports payment or honoraria for advisory boards or lectures for Agios Pharmaceuticals, and Chiesi. LC, JLD, JHE, EF, SG, GMP, KO, HS, and KU are employees and shareholders of Agios Pharmaceuticals. AG reports consultancy fees from Agios Pharmaceuticals, Novo Nordisk, and Pharmacosmos; participation on a data safety monitoring board or advisory board for Disc Pharmaceuticals, Novo Nordisk, and Vertex Pharmaceuticals; and research funding from Agios Pharmaceuticals, Bristol Myers Squibb, Novo Nordisk, and Sanofi. KMM reports consultancy fees from Agios Pharmaceuticals, Bristol Myers Squibb (Celgene), CRISPR Therapeutics, Novartis, Novo Nordisk, Pharmacosmos, and Vifor; and research funding from Pharmacosmos. JBP reports advisory board membership for Agios Pharmaceuticals, Bluebird Bio, Bristol Myers Squibb, Silence, and Vifor; and research funding from Silence. EJvB reports consulting fees and research funding from Agios Pharmaceuticals. VV reports research funding from Agios Pharmaceuticals, Bristol Myers Squibb (Celgene), DisperSol Technologies, Ionis Pharmaceuticals, Novartis, and Pharmacosmos, The Government Pharmaceutical Organization, and Vifor; and consulting fees from Agios Pharmaceuticals, Bristol Myers Squibb (Celgene), DisperSol Technologies, Ionis Pharmaceuticals, Novartis, Pharmacosmos, and Vifor. KHKM reports consultancy fees from Agios Pharmaceuticals, Alexion Pharmaceuticals, Biossil, Bristol Myers Squibb, Forma Therapeutics, Novo Nordisk, Pfizer, and Vertex

Pharmaceuticals; honoraria from Agios Pharmaceuticals and Bristol Myers Squibb; membership on the Board of Directors and advisory committees for Sangamo; and research funding from Agios Pharmaceuticals and Pfizer. AK reports advisory board fees from Agios Pharmaceuticals, Bristol Myers Squibb, Novo Nordisk, Pfizer, and Vertex; and honoraria from Agios Pharmaceuticals, Bristol Myers Squibb, and Vertex. PG declares no competing interests.

Data sharing

Qualified researchers can request access to related clinical study documents from datasharing@agios.com. The following considerations will be taken into account as part of the review: ability for external researchers to re-identify trial participants, such as small rare disease trials or single-centre trials; language used in data and requested documents (eg, English or other); informed consent language with respect to allowance for data sharing; plan to re-evaluate safety or efficacy data summarised in the approved product labelling; and potential conflict of interest or competitive risk.

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